

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
 mendamar.badrall@gmail.com |  +36 70 7385 744 |  Budapest, Hungary

I'm a diligent and hard-working student with a passion for applying computational thinking to the most interesting problems. I aspire to contribute to the advancements in scientific discovery, machine learning and computer vision. I'm an avid Julia enthusiast and Python programmer.

EDUCATION

Budapest University of Technology and Economics (BME) M.S. in Computer Science Engineering <i>Focus: Data Science & Artificial Intelligence</i>	Sep, 2025 – June, 2027 Budapest, Hungary
National University of Mongolia (NUM) B.S. in Applied Mathematics Thesis: Evaluating land degradation using image processing methods	Sep, 2021 – June, 2025 Ulaanbaatar, Mongolia

EXPERIENCE

Center of Mathematics Application Research Laboratory <i>Research Undergraduate Intern</i>	Oct, 2023 – June, 2025
– Lab Leadership: Selected as Undergraduate Lab Lead in final year; coordinated weekly research progress meetings and facilitated communication between students and faculty. – Infrastructure Management: Managed and maintained the lab computing resources in the Linux environment. – Research: Conducted research on image processing techniques to quantify plant types from drone imagery using segmentation methods (thresholding, K-means, Moran's I). – Documentation: Authored a comprehensive laboratory handbook and technical documentation to standardize onboarding for future researchers.	

PROJECTS

[Deep Learning Foundations and Concepts Book Figures Reproduction](#)

- Self-initiated project to reproduce key examples from C.Bishop and H.Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of **polynomial regression** and **probability theory** based on early chapters of the book.
- Published reproducible Jupyter and Pluto.jl notebooks on GitHub, emphasizing clear explanation, interactivity, and visualization.

SKILLS

Programming Languages:	Python, Julia, C, C++, R
ML Frameworks:	PyTorch
Markup & Docs:	Markdown, Quarto, L ^A T _E X
Databases & Tools:	SQLite, OpenCV, Git, Vim
Theoretical Background:	Applied Mathematics, Digital Image Processing